

Unit 9 Guided Notes — Volume of 3D Solids

Cylinders · Cones · Spheres. $V = \pi r^2 h$, $V = \frac{1}{3}\pi r^2 h$, $V = \frac{4}{3}\pi r^3$.

Standard: NC.8.G.9

 **Blank Worksheet (PDF)**

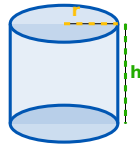
 **Answer Key (PDF)**

 **EC/Modified Notes (PDF)**

All volume formulas use π (pi ≈ 3.14). Square the radius (r^2) for cylinders and cones — CUBE it (r^3) for spheres. Always include CUBIC units (cm^3 , in^3 , ft^3).

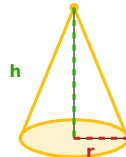
1 Key Formulas

Cylinder



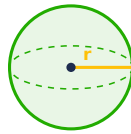
$$V = \pi r^2 h$$

Cone



$$V = \frac{1}{3}\pi r^2 h$$

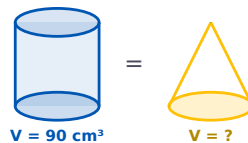
Sphere



$$V = \frac{4}{3}\pi r^3$$

🔑 **THE BIG RELATIONSHIP:** A cone is exactly $\frac{1}{3}$ of a cylinder with the same radius and height.

Quick Check

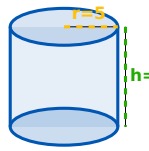


MC The cylinder above holds 90 cm³. The cone has the SAME radius and height. What's the cone's volume?

✓ A cone is always $\frac{1}{3}$ of the matching cylinder: $90 \div 3 = 30 \text{ cm}^3$.

2 Worked Examples

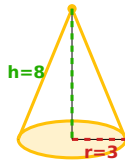
Cylinder: $r = 5 \text{ cm}$, $h = 10 \text{ cm}$



Substitute into $V = \pi r^2 h$: $V = \pi(5)^2(10) = \pi(25)(10)$

✓ Answer: $V = 250\pi \approx 785.4 \text{ cm}^3$

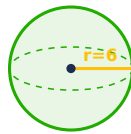
💡 Cone: $r = 3 \text{ in}$, $h = 8 \text{ in}$



Substitute into $V = \frac{1}{3}\pi r^2 h$: $V = \frac{1}{3} \cdot \pi(3)^2(8) = \frac{1}{3} \cdot 72\pi$

✓ Answer: $V = 24\pi \approx 75.4 \text{ in}^3$

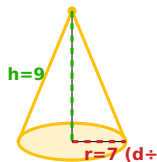
💡 Sphere: $r = 6 \text{ ft}$



Substitute into $V = \frac{4}{3}\pi r^3$: $V = \frac{4}{3} \cdot \pi(6)^3 = \frac{4}{3} \cdot 216\pi$

✓ Answer: $V = 288\pi \approx 904.3 \text{ ft}^3$

💡 Cone with Diameter Given: diameter = 14 cm, $h = 9 \text{ cm}$



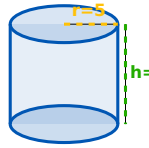
Diameter 14 → radius: $r = 7$

Substitute: $V = \frac{1}{3}\pi(7^2)(9) = \frac{1}{3}(441\pi)$

✓ Answer: $V = 147\pi \approx 461.8 \text{ cm}^3$

3 Error Analysis

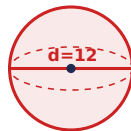
SPOT THE ERROR Cylinder: $r = 5$, $h = 10$.



A student wrote $V = \pi(5)(10) = 50\pi \approx 157.1$.

Forgot to SQUARE the radius! $V = \pi(5^2)(10) = 250\pi \approx 785.4$. The formula is $\pi r^2 h$.

SPOT THE ERROR Sphere: diameter = 12.



A student wrote $r = 12$, $V = \frac{4}{3}\pi(12^3) \approx 7238.2$.

Used DIAMETER instead of radius! $r = 6$, not 12. $V = \frac{4}{3}\pi(6^3) = 288\pi \approx 904.8$. Diameter $\div 2$ = radius.

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[← Back to Unit 9](#)